V4.7_CHECK Comprehensive Statistical Summary

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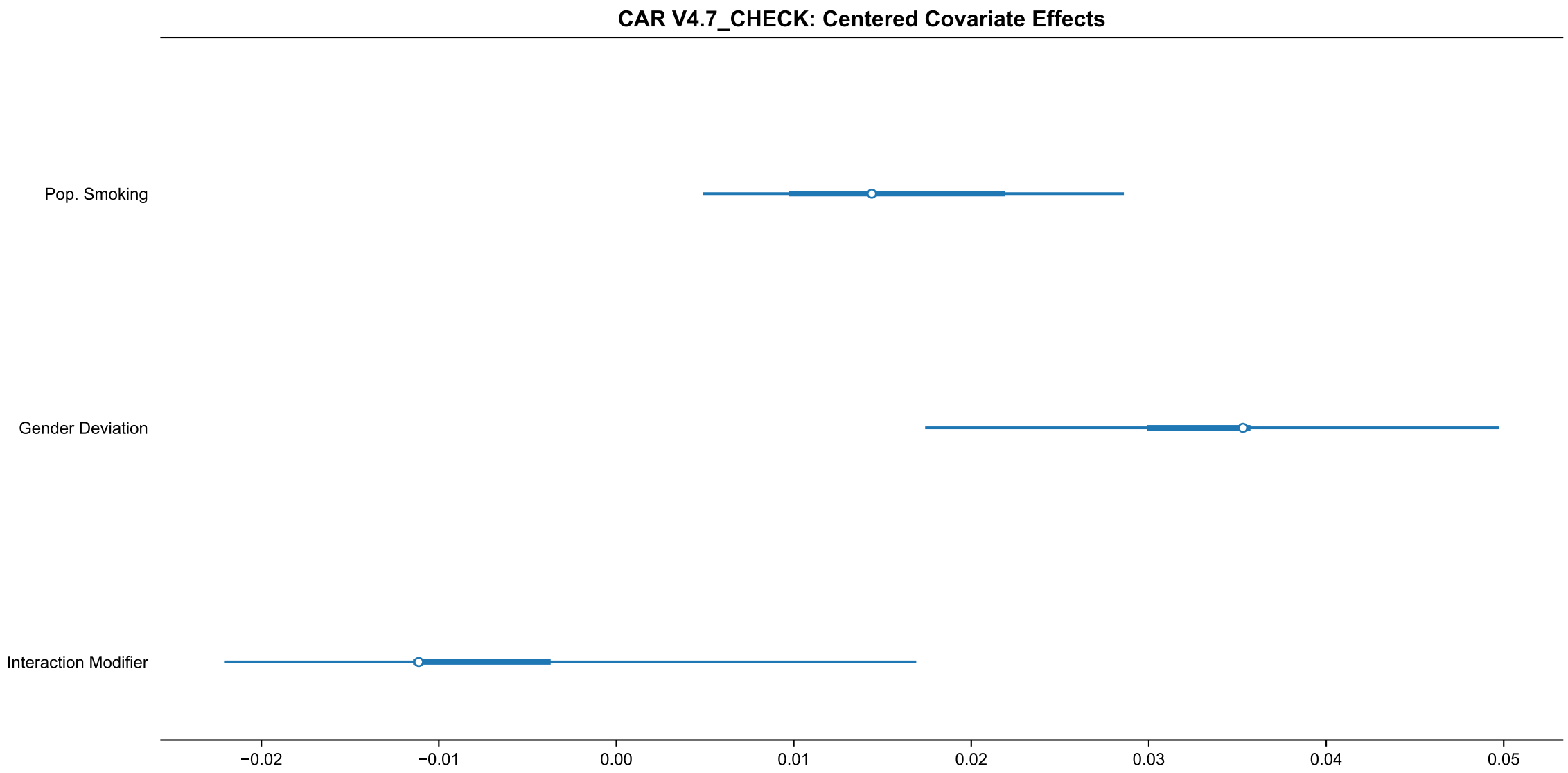
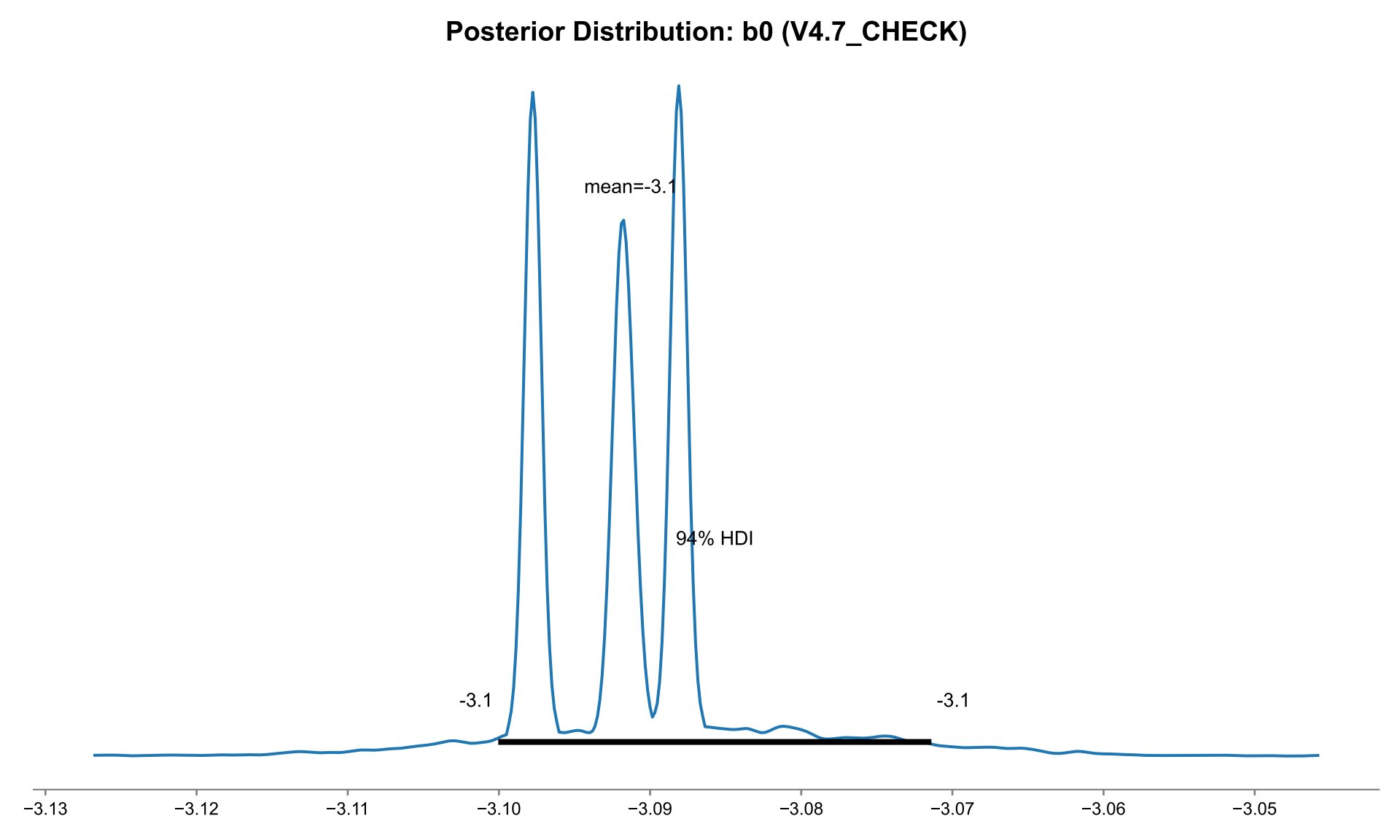
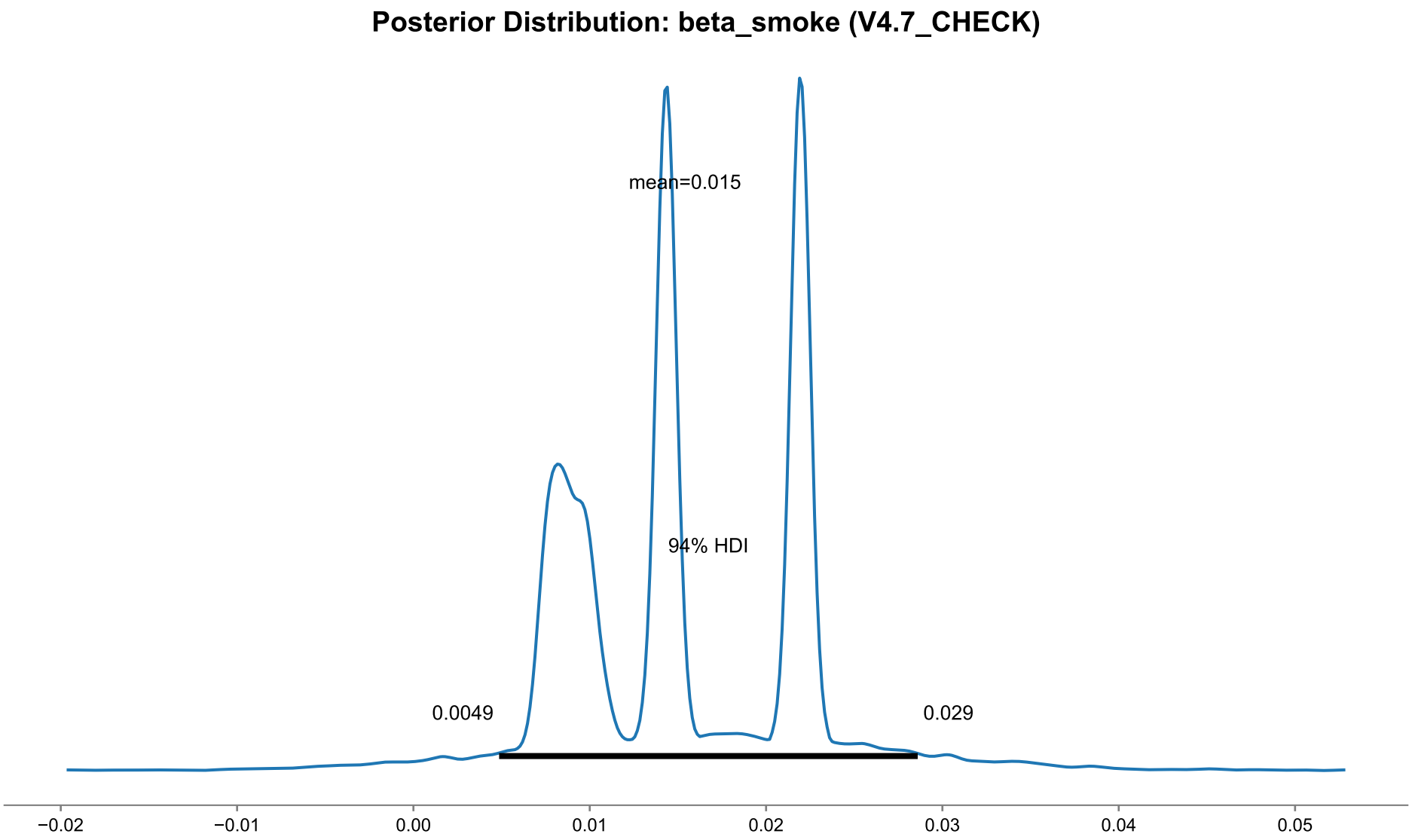
CAR V4.7_CHECK MODEL AUDIT | car_model_report_v4.7_check
================================================================
WAIC (Deviance): 4204.43 | LOO-CV: 4402.90
Divergences: 0 | Max R-hat: 2.150

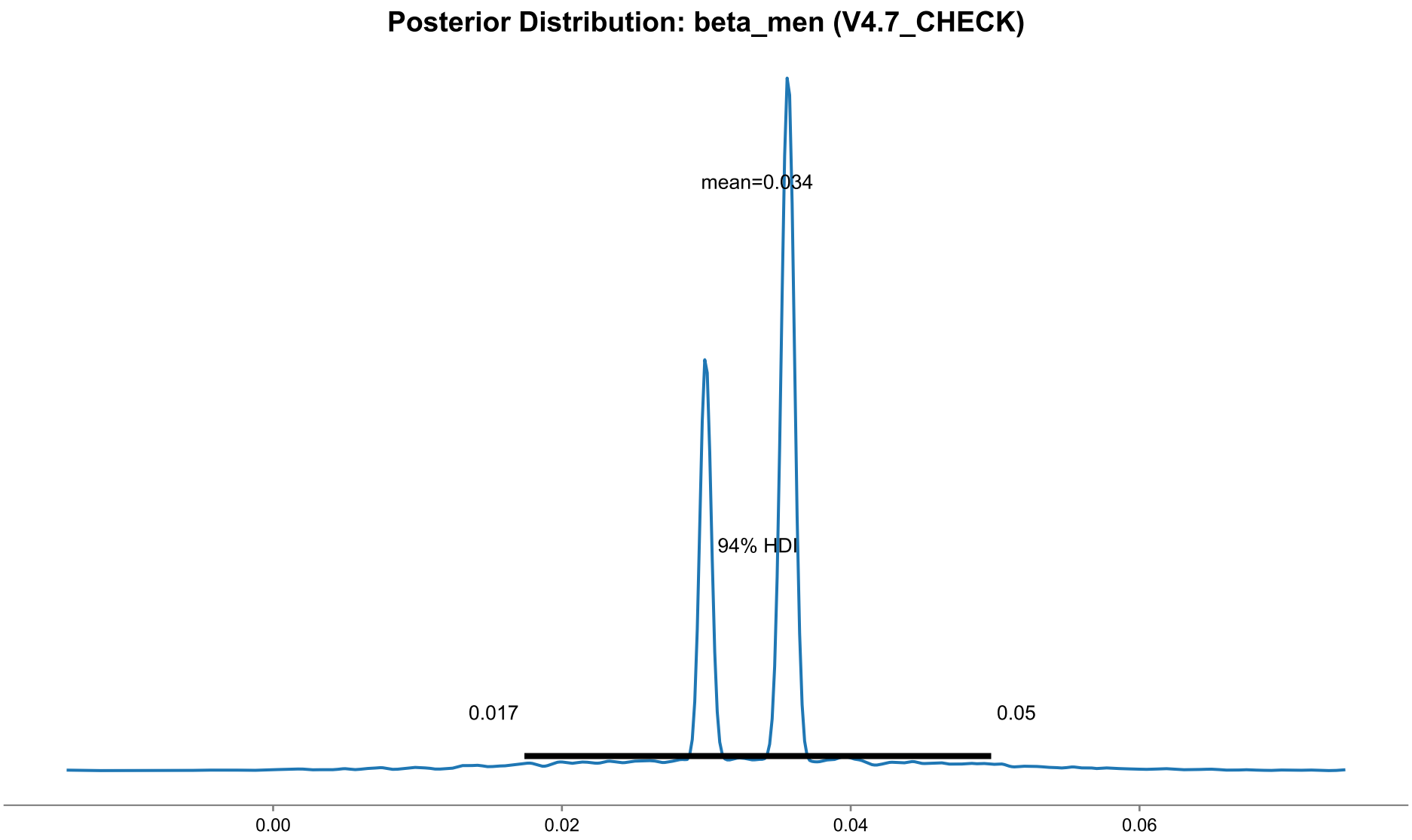
PUBLIC HEALTH INTERPRETATION:
- Pop. Smoking Effect: +1.51% risk per 5.3% increase
- Gender Modifier (Men): -0.70% additional risk adjustment

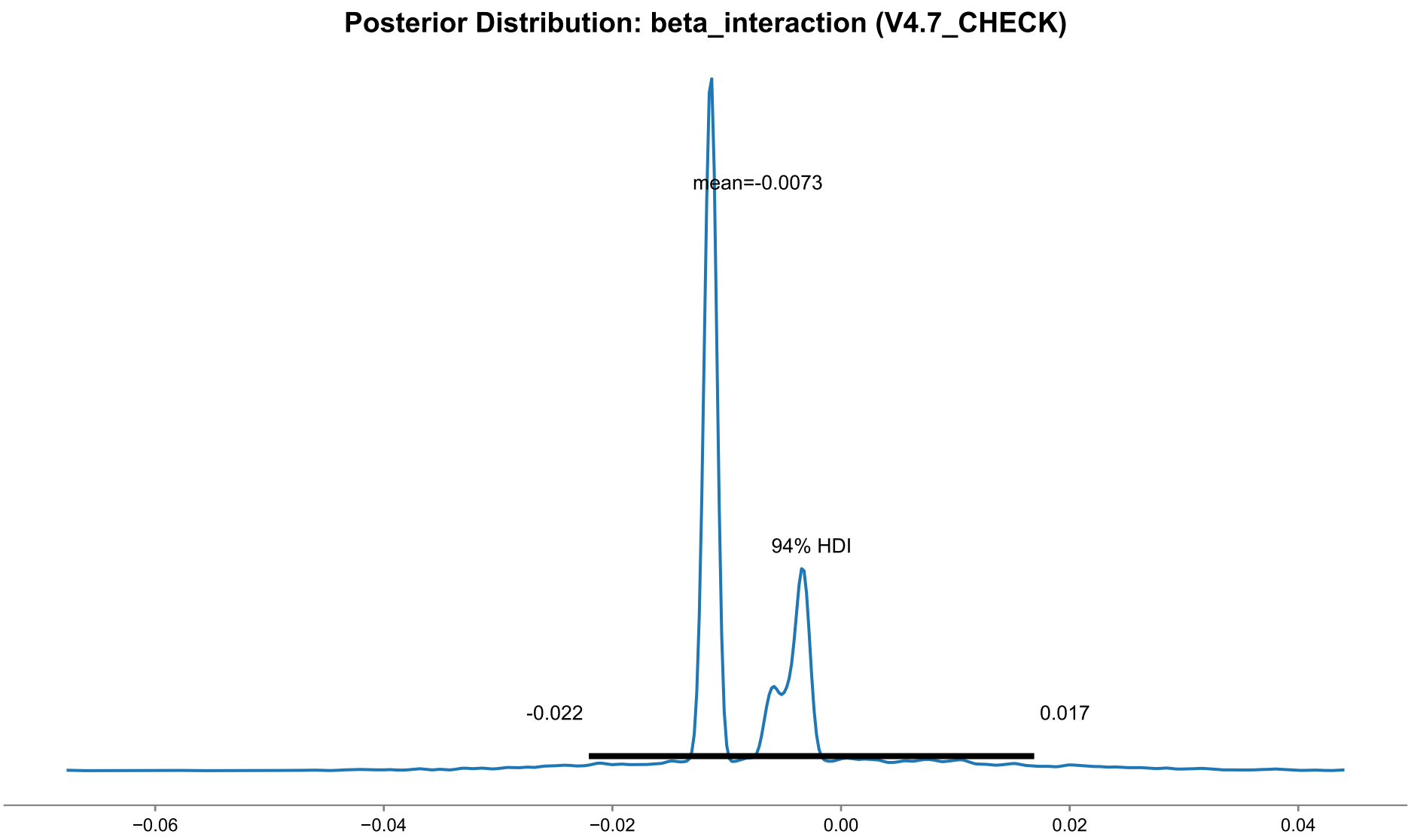
                    mean     sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  ess_bulk  ess_tail  r_hat
b0                -3.091  0.007  -3.100   -3.071      0.002    0.002      10.0      26.0   1.85
beta_smoke         0.015  0.007   0.005    0.029      0.002    0.001       8.0      27.0   1.75
beta_men           0.034  0.007   0.017    0.050      0.001    0.003      16.0      27.0   1.90
beta_interaction -0.007  0.009  -0.022    0.017      0.002    0.003      11.0      28.0   2.15
sigma              0.232  0.015   0.210    0.252      0.006    0.003       6.0      16.0   1.79
rho                0.358  0.070   0.239    0.483      0.023    0.015       9.0      28.0   2.00
```

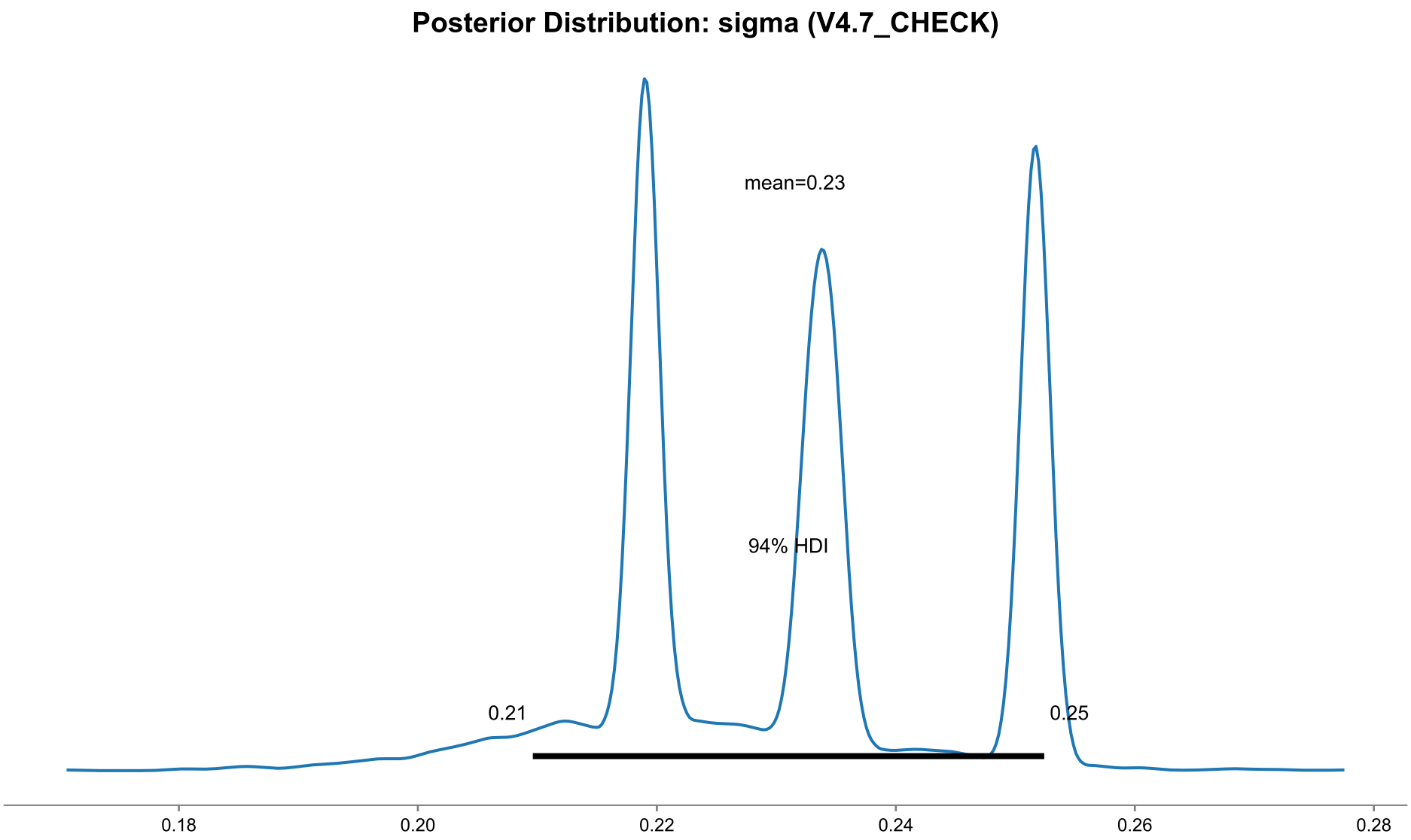

CAR V4.7_CHECK: Centered Covariate Effects
Pop. Smoking
Gender Deviation
Interaction Modifier
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
0.05

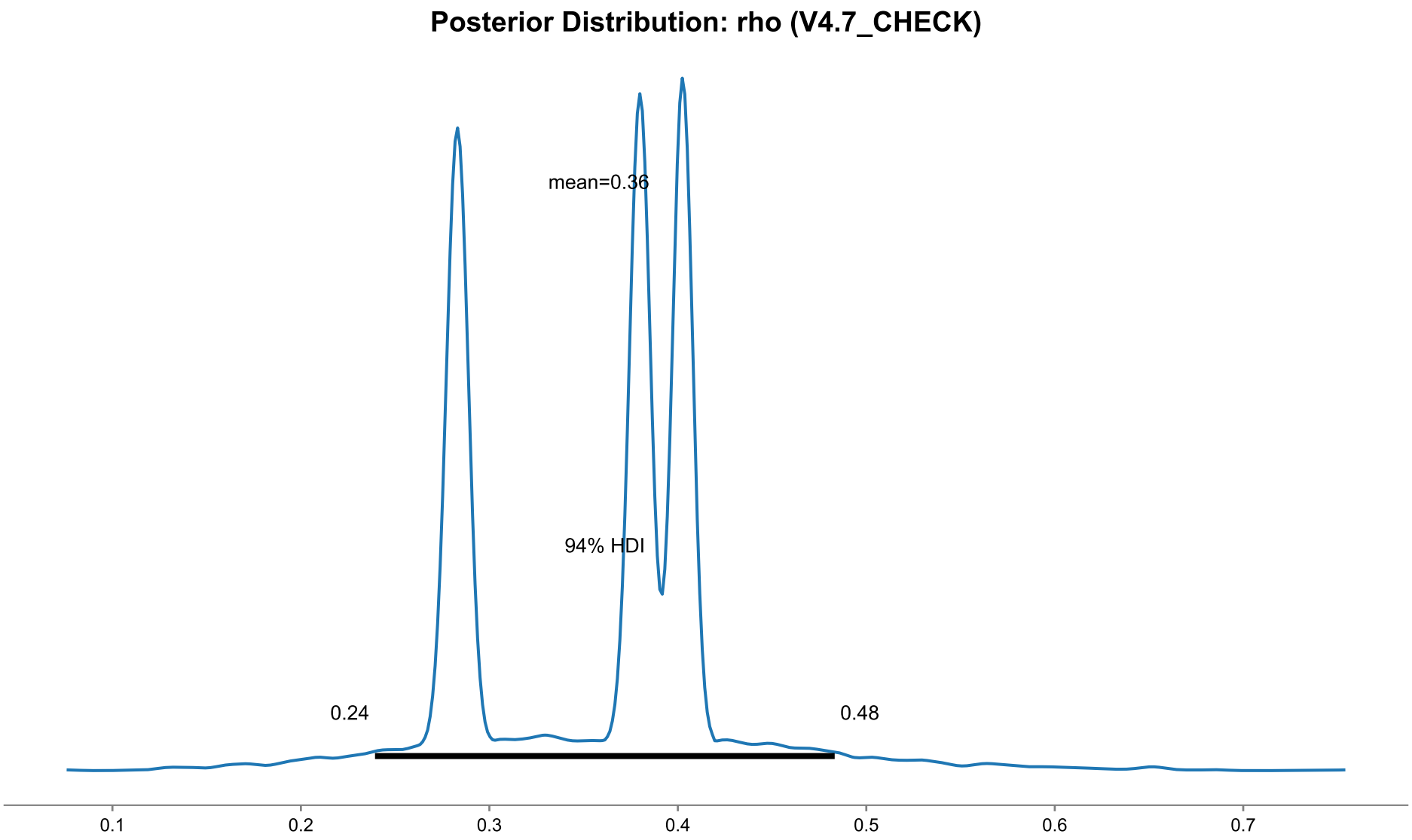
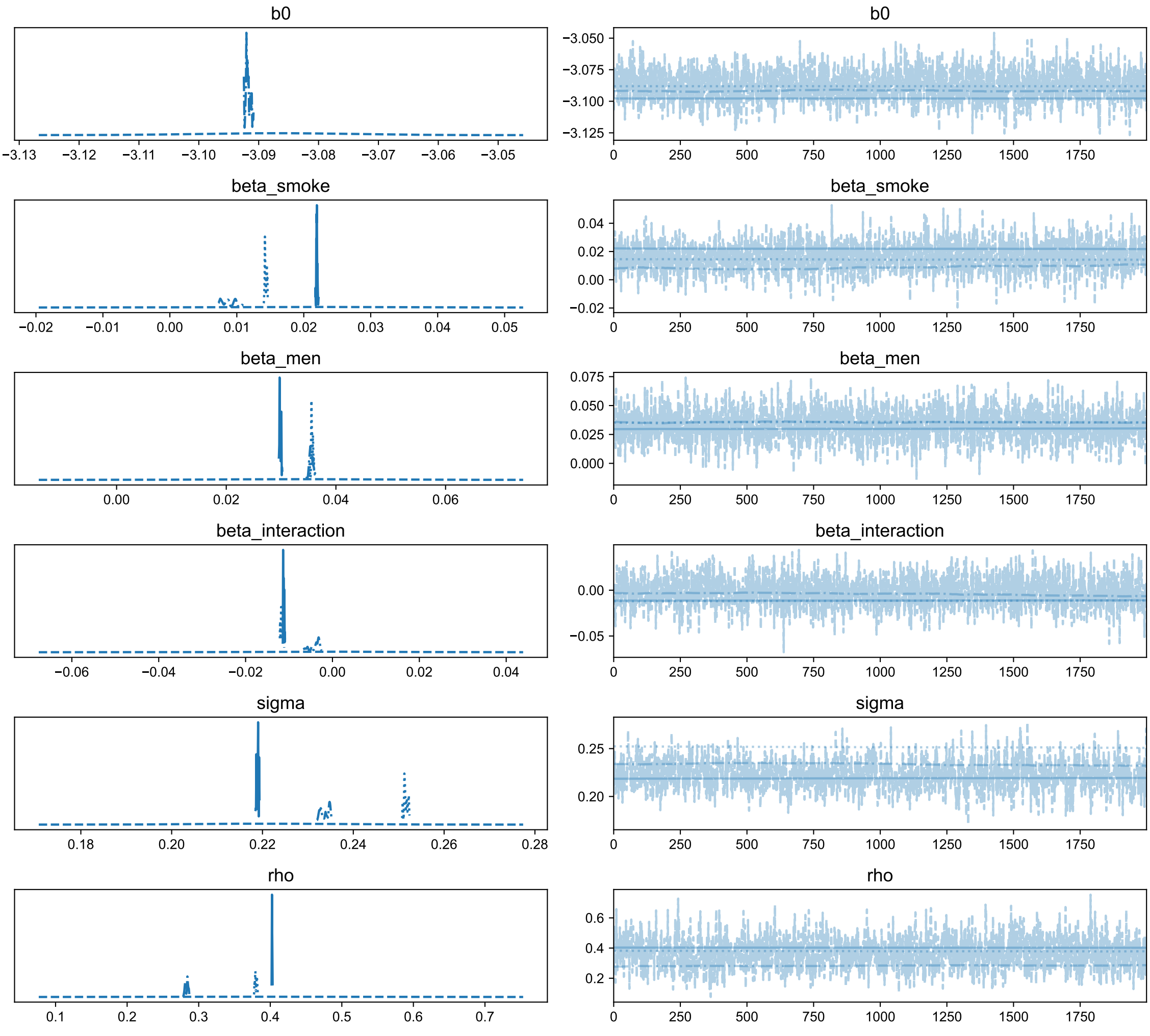

Posterior Distribution: b0 (V4.7_CHECK)
mean=-3.1
94% HDI
-3.1
-3.1
−3.13
−3.12
−3.11
−3.10
−3.09
−3.08
−3.07
−3.06
−3.05


Posterior Distribution: beta_smoke (V4.7_CHECK)
mean=0.015
94% HDI
0.0049
0.029
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
0.05


Posterior Distribution: beta_men (V4.7_CHECK)
mean=0.034
94% HDI
0.017
0.05
0.00
0.02
0.04
0.06


Posterior Distribution: beta_interaction (V4.7_CHECK)
mean=-0.0073
94% HDI
-0.022
0.017
−0.06
−0.04
−0.02
0.00
0.02
0.04


Posterior Distribution: sigma (V4.7_CHECK)
mean=0.23
94% HDI
0.21
0.25
0.18
0.20
0.22
0.24
0.26
0.28


Posterior Distribution: rho (V4.7_CHECK)
mean=0.36
94% HDI
0.24
0.48
0.1
0.2
0.3
0.4
0.5
0.6
0.7

V4.7_CHECK MCMC Chain Traces